

TITLE: Case Study-Target SQL

Question 1: Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table.

Answer:

```
select column_name,data_type
from `target-cs-415302.target.INFORMATION_SCHEMA.COLUMNS`
where table_name = 'customers';
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON
Row	data_type ▼	column_name ▼		
1	STRING	customer_id		
2	STRING	customer_unique_id		
3	INT64	customer_zip_code_prefix		
4	STRING	customer_city		
5	STRING	customer_state		

Insights:

All columns in this example are saved as strings (**VARCHAR**), except customer_zip_code_prefix it is stored as **integers**. This shows that most of the data are string in nature

2. Get the time range between which the orders were placed.

Answer:

```
select min(order_purchase_timestamp) as min_order_timestamp,
max(order_purchase_timestamp) as max_order_timestamp
from `target-cs-415302.target.orders`
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	I
Row	min_order_timestamp ▼	max_order_timestamp ▼			
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC			

Insights:

The timestamp will be useful for analyze overall order patterns over a certain time, it can be helpful to know the time range of the orders.

3. Count the Cities & States of customers who ordered during the given period.

Answer:

```
select count(distinct customer_city) as cty,
count(distinct customer_state) as state
from `target-cs-415302.target.customers`
where customer_id in
(select distinct customer_id
from `target-cs-415302.target.orders`
where order_purchase_timestamp between '2016-09-04' and '2018-10-17')
```

Screenshot:

JOB INFORMATION		RESULTS	CHART
Row	cty ▼	state ▼	
1	4119	27	

Insights:

The dataset consists of 27 states and 4119 cities, providing insight into the geographic distribution of our clientele. Analyzing these figures can reveal the diversity or concentration of our clients in different regions, for regional analysis, identifying hotspots.

Question 2: In-depth Exploration

1. Is there a growing trend in the no. of orders placed over the past years?

Answer:

```
select extract(year from order_purchase_timestamp) as year,
count(*) as order_count
from `target-cs-415302.target.orders`
group by 1
order by 1
```

Screenshot:

JOB INFORMATION		RESULTS	CHART
Row	year ▼	order_count ▼	
1	2016	329	
2	2017	45101	
3	2018	54011	

Insights:

There has been an upward trend in the number of orders over the past few years after examining the results. A favorable trend can be seen if the order number regularly rises in year over year.

2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Answer:

```
select extract(year from order_purchase_timestamp) as year,
extract (month from order_purchase_timestamp ) as month,
count(*) as order_count
from `target-cs-415302.target.orders`
group by 1,2
order by year asc,month asc;
```

Screenshot:

Query results					SAVE RESULTS ▼	EX
JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	year ▼	month ▼	order_count ▼			
1	2016	9	4			
2	2016	10	324			
3	2016	12	1			
4	2017	1	800			
5	2017	2	1780			
6	2017	3	2682			
7	2017	4	2404			
8	2017	5	3700			
9	2017	6	3245			
10	2017	7	4026			

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Insights:

Understanding monthly seasonality can help with planning of operations, marketing tactics, and consumer behavior. It can be used in better planning of promotional activities, inventory management optimization.

--3. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

-- 0-6 hrs: Dawn -- 7-12 hrs: Mornings -- 13-18 hrs: Afternoon -- 19-23 hrs: Night

Answer:

```
select case
when extract (HOUR from order_purchase_timestamp) between 0 and
6 then 'Dawn'
when extract (HOUR from order_purchase_timestamp) between 7 and 12
then 'Morning'
when extract (HOUR from order_purchase_timestamp) between 13 and 18
then 'Afternoon'
when extract (HOUR from order_purchase_timestamp) between 19 and 23
then 'Night'
end order_time_interval,
count(*) as order_count
from `target-cs-415302.target.orders`
group by order_time_interval
order by order_count desc;
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON
Row	order_time_interval	order_count		
1	Afternoon	38135		
2	Night	28331		
3	Morning	27733		
4	Dawn	5242		

Insights:

The query divides order timestamps into four time groups (Dawn, Morning, Afternoon, Night) and sorts results based on orders. Analyzing Brazilian clients' ordering habits helps identify their preferences. For example, afternoons are the busiest online shopping periods, suggesting efficient operations. Additionally, customers buy least during dawn.

Question 3: Evolution of E-commerce orders in the Brazil region

--1. Get the month-on-month no. of orders placed in each state—

Answer:

```
select extract(month from order_purchase_timestamp) as order_month,
count(*) as no_of_orders,
c.customer_state
from `target-cs-415302.target.orders` o
join `target-cs-415302.target.customers` c
on o.customer_id= c.customer_id
group by order_month,c.customer_state
order by order_month,c.customer_state;
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	order_month	no_of_orders	customer_state			
1	1	8	AC			
2	1	39	AL			
3	1	12	AM			
4	1	11	AP			
5	1	264	BA			
6	1	99	CE			
7	1	151	DF			
8	1	159	ES			
9	1	164	GO			
10	1	66	MA			
11	1	971	MG			

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Insights:
Analyzing the query results allows us to identify trends in order volumes across states, identifying high order volumes and significant changes. The data reveals SP has the highest order volume every month, allowing us to target marketing efforts, identify operational issues, and optimize inventory management.

-- 2. How are the customers distributed across all the states?—

Answer:

```
select customer_state, count(distinct customer_id) as total_customers
from `target-cs-415302.target.customers`
group by 1
order by 2 desc;
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	customer_state ▼	total_customers ▼				
1	SP	41746				
2	RJ	12852				
3	MG	11635				
4	RS	5466				
5	PR	5045				
6	SC	3637				
7	BA	3380				
8	DF	2140				
9	ES	2033				
10	GO	2020				

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Insights:

Analyzing the query results reveals the distribution of clients across states, with SP having the highest client base and RR having the fewest. This information aids in market targeting, expansion opportunities, and optimizing company strategy.

Question 4: Impact on Economy: Analyze the money movement by ecommerce by looking at order prices, freight, and others.

----1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only). You can use the "payment_value" column in the payments table to get the cost of orders—

Answer:

```
select ( sum(case when extract(YEAR from o.order_purchase_timestamp) = 2018
and extract(MONTH from o.order_purchase_timestamp) between 1 and 8 then
p.payment_value else 0 end) -
```

```
sum(case when extract(YEAR from o.order_purchase_timestamp) = 2017
and extract(MONTH from o.order_purchase_timestamp) between 1 and 8 then
p.payment_value else 0 end) ) /
```

```
sum(case when extract(YEAR from o.order_purchase_timestamp) = 2017
and extract(MONTH from o.order_purchase_timestamp) between 1 and 8 then
```

```

p.payment_value else 0 end) * 100 as cost_increase_percentage
from `target-cs-415302.target.payments` as p
join `target-cs-415302.target.orders` as o
on p.order_id=o.order_id;

```

Screenshot:

JOB INFORMATION		RESULTS	CHART
Row		cost_increase_percentage	
1		136.97687164665447	

Insights:

Orders from January to August in 2017 and 2018 showed a growth rate of approximately 137%, as analyzed by monthly prices between the years.

--2. Calculate the Total & Average value of order price for each state—

Answer:

```

select c.customer_state,
sum(p.payment_value) as total_price,
avg(p.payment_value) as avg_price
from `target-cs-415302.target.payments` p
join `target-cs-415302.target.orders` o
on p.order_id = o.order_id
join `target-cs-415302.target.customers` c
on c.customer_id = o.customer_id
group by 1
order by 1

```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	customer_state	total_price	avg_price			
1	AC	19680.62000000...	234.2930952380...			
2	AL	96962.06	227.0774238875...			
3	AM	27966.93	181.6034415584...			
4	AP	16262.80000000...	232.3257142857...			
5	BA	616645.8200000...	170.8160166204...			
6	CE	279464.0299999...	199.9027396280...			
7	DF	355141.08	161.1347912885...			
8	ES	325967.55	154.7069530137...			
9	GO	350092.3099999...	165.7634043560...			
10	MA	152523.02	198.8566101694...			
11	MG	1872257.259999...	154.7064336473...			

Insights:

The "total_price" column displays the total order prices for each state, while the "avg_price" column shows the normal order value and average order price. Analyzing these results can identify profitable marketplaces, aid in developing marketing strategies, and consider the context of each state for informed decision-making.

--3.Calculate the Total & Average value of order freight for each state--

```
select c.customer_state,
sum(i.freight_value)as total_freight,
avg(i.freight_value) as avg_freight
from `target-cs-415302.target.orders` o
join `target-cs-415302.target.order_items` i
on i.order_id = o.order_id
join `target-cs-415302.target.customers` c
on c.customer_id = o.customer_id
group by 1
order by c.customer_state
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	customer_state	total_freight	avg_freight			
1	AC	3686.749999999...	40.07336956521...			
2	AL	15914.589999999...	35.84367117117...			
3	AM	5478.889999999...	33.20539393939...			
4	AP	2788.500000000...	34.00609756097...			
5	BA	100156.6799999...	26.36395893656...			
6	CE	48351.589999999...	32.71420162381...			
7	DF	50625.499999999...	21.04135494596...			
8	ES	49764.599999999...	22.05877659574...			
9	GO	53114.979999999...	22.76681525932...			
10	MA	31523.770000000...	38.25700242718...			
11	MG	270853.4600000...	20.63016680630...			

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Insights:

We can find states with high total freight costs,by analyzing the results, which could point to regions with higher shipping prices or logistical difficulties.Understanding the differences in order freight rates between states can offer information about local shipping habits, supplier locations.

Question 5. Analysis based on sales, freight, and delivery time.

--1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order. Do this in a single query. You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula: $\text{time_to_deliver} = \text{order_delivered_customer_date} - \text{order_purchase_timestamp}$ $\text{diff_estimated_delivery} = \text{order_estimated_delivery_date} - \text{order_delivered_customer_date}$ —

Answer:

```
select order_id,  
datetime_diff(order_delivered_customer_date,order_purchase_timestamp,day) as  
delivery_time_in_days,  
datetime_diff(order_estimated_delivery_date,order_purchase_timestamp,day) as  
different_est_delivery  
from `target-cs-415302.target.orders`  
where order_status = 'delivered'  
order by order_id desc
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	order_id	delivery_time_in_days		different_est_delivery		
1	fffe41c64501cc87c801fd61db...	5		18		
2	fffe18544ffabc95dfada21779c...	1		10		
3	fffce4705a9662cd70adb13d4a...	4		17		
4	ffcd46ef2263f404302a634eb...	9		17		
5	fffc94f6ce00a00581880bf54a...	17		24		
6	fffbce3b5462987e66fb49b1c5...	16		33		
7	fffb9224b6fc7c43ebb0904318...	21		30		
8	fffb2ef8874127f75b52b64388...	17		27		
9	fffb0b1a50e65c449020434fa8...	34		32		
10	fffa82886406ccf10c7b4e35c4f...	21		36		
11	fff90cdcb3b2e6cfb397d05d56...	4		13		

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Insights:

Analyzing the “delivery_time_in_days” and “different_est_delivery” columns can provide insights into the effectiveness of the delivery process, including delays or early deliveries. These insights can identify trends, elements affecting delivery times, enabling better customer management, and optimization of logistics operations.

--2. Find out the top 5 states with the highest & lowest average freight value—

Answer:

```
SELECT
  high.customer_state AS high_state,
  high.average_freight_value AS high_avg_freight,
  low.customer_state AS low_state,
  low.average_freight_value AS low_avg_freight
FROM
(
  SELECT
    c.customer_state,
    ROUND(AVG(p.freight_value),2) AS average_freight_value,
    ROW_NUMBER() OVER(ORDER BY
(ROUND(AVG(p.freight_value),2))DESC) AS row_val_1
  FROM `target-cs-415302.target.orders` AS o
  JOIN `target-cs-415302.target.order_items` AS p
  ON o.order_id = p.order_id
  JOIN `target-cs-415302.target.customers` AS c
  ON o.customer_id = c.customer_id
  GROUP BY
  c.customer_state
  ORDER BY
  average_freight_value DESC
  LIMIT
  5
) AS high
JOIN
(
  SELECT
    c.customer_state,
    ROUND(AVG(p.freight_value),2) AS average_freight_value,
    ROW_NUMBER() OVER(ORDER BY (ROUND(AVG(p.freight_value),2))) AS
row_val_2
  FROM `target-cs-415302.target.orders` AS o
  JOIN `target-cs-415302.target.order_items` AS p
  ON o.order_id = p.order_id
  JOIN `target-cs-415302.target.customers` AS c
  ON o.customer_id = c.customer_id
  GROUP BY
  c.customer_state
  ORDER BY
```

```

average_freight_value
LIMIT
5
) AS low
ON high.row_val_1 = low.row_val_2;

```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	high_state	high_avg_freight	low_state	low_avg_freight		
1	RR	42.98	SP	15.15		
2	PB	42.72	PR	20.53		
3	RO	41.07	MG	20.63		
4	AC	40.07	RJ	20.96		
5	PI	39.15	DF	21.04		

Insights:

States with high average freight values, such as RR and PB, may face higher shipping prices due to remote locations, higher transportation costs, or supply chain issues. Companies can optimize logistics operations by locating places with lower shipping prices, such as SP and PR. This data can help develop focused initiatives, bargain freight costs, and identify cost reduction opportunities.

--3. Find out the top 5 states with the highest & lowest average delivery time--

Answer:

```

with cte as
(select
  c.customer_state,
  round(avg(t1.delivery_time),2) as avg_delivery_time
from
(select
  *,
  timestamp_diff(order_delivered_customer_date,
  order_purchase_timestamp, day) AS delivery_time,
from `target-cs-415302.target.orders`
where
  order_status = 'delivered' and
  order_delivered_customer_date is not null
  order by order_purchase_timestamp)as t1

join

`target-cs-415302.target.customers` as c
on t1.customer_id = c.customer_id

```

```

group by c.customer_state
order by avg_delivery_time)
select
  c1.customer_state as low_state,
  c1.avg_delivery_time as low_avg_delivery_time,
  c2.customer_state as high_state,
  c2.avg_delivery_time as high_avg_delivery_time
from
(select
  *,
  row_number() over(order by cte.avg_delivery_time desc) as row_val_2
from
  cte
ORDER BY row_val_2) AS c2

JOIN

(SELECT
  *,
  row_number() over (order by cte.avg_delivery_time)as
row_val_1
from
  cte
order by row_val_1) AS c1
ON c1.row_val_1 = c2.row_val_2
LIMIT 5;

```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	low_state ▾	low_avg_delivery_time	high_state ▾	high_avg_delivery_time		
1	SP	8.3	RR	28.98		
2	PR	11.53	AP	26.73		
3	MG	11.54	AM	25.99		
4	DF	12.51	AL	24.04		
5	SC	14.48	PA	23.32		

Insights:

The study identifies areas with efficient delivery operations, faster transit times, and solid logistics networks in states like RR and AP, with high average delivery times. These insights can help improve customer satisfaction, operational efficiency, and delivery process optimization. This helps the company focus on areas for improvement and in enhancing customer experiences.

--4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery. You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state--

Answer:

```
WITH delivery_speed AS (  
  SELECT  
    c.customer_state,  
    AVG(DATE_DIFF(o.order_delivered_customer_date,  
o.order_estimated_delivery_date, DAY)) AS avg_delivery_speed,  
    ROW_NUMBER() OVER (ORDER BY  
AVG(DATE_DIFF(o.order_delivered_customer_date,  
o.order_estimated_delivery_date, DAY))) AS rank_fastest  
  FROM `target-cs-415302.target.orders` AS o  
  JOIN `target-cs-415302.target.customers` AS c  
  ON o.customer_id = c.customer_id  
  WHERE o.order_delivered_customer_date IS NOT NULL AND  
o.order_estimated_delivery_date IS NOT NULL  
  GROUP BY c.customer_state  
)  
SELECT customer_state, avg_delivery_speed  
FROM delivery_speed  
WHERE rank_fastest <= 5  
ORDER BY avg_delivery_speed;
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON
Row	customer_state	avg_delivery_speed		
1	AC	-19.762500000000006		
2	RO	-19.13168724279836		
3	AP	-18.731343283582088		
4	AM	-18.60689655172413		
5	RR	-16.414634146341463		

Insights:

Our company, based in AC, RO, AP, and AM states, can leverage the highest average delivery speed to attract more clients and enhance customer satisfaction. This data can be used to improve operations, expansion opportunities in these areas.

Question 6: Analysis based on the payments

--1. Find the month-on-month no. of orders placed using different payment types—

Answer:

```
select
format_timestamp('%Y-%m', o.order_purchase_timestamp) as
month,p.payment_type,
count(distinct o.order_id) as order_cnt
from `target-cs-415302.target.orders` as o
join `target-cs-415302.target.payments` as p
on o.order_id = p.order_id
group by month, p.payment_type
order by month
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	month	payment_type	order_cnt			
1	2016-09	credit_card	3			
2	2016-10	credit_card	253			
3	2016-10	UPI	63			
4	2016-10	voucher	11			
5	2016-10	debit_card	2			
6	2016-12	credit_card	1			
7	2017-01	credit_card	582			
8	2017-01	UPI	197			
9	2017-01	voucher	33			
10	2017-01	debit_card	9			
11	2017-02	credit_card	1347			

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Insights:

Credit card usage was highest in November 2017. Tracking order counts can help analyze peak months, and consumer behavior effects. This information can help to optimize payment procedures and enhance customer experiences.

--2. Find the no. of orders placed on the basis of the payment installments that have been paid--

Answer:

```
select payment_installments, count(distinct order_id) as order_count
from `target-cs-415302.target.payments`
group by 1
ORDER BY 1
```

Screenshot:

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	payment_installment	order_count				
1	0	2				
2	1	49060				
3	2	12389				
4	3	10443				
5	4	7088				
6	5	5234				
7	6	3916				
8	7	1623				
9	8	4253				
10	9	644				
11	10	5315				

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Insights:

The analysis of 49060 orders with payment installment 1 can reveal customer preferences for budgeting and financing. The distribution of orders based on payment installments can reveal buying habits, indicating the popularity of payment installment alternatives.